17A 1 TRANSITION METAL ELECTRONIC CONFIGURATIONS

SPECIFICATION REFERENCE
17.1 17.2 17.3

LEARNING OBJECTIVES

- Know that transition metals are d-block elements that form one or more stable ions with incompletely filled d-orbitals.
- Be able to deduce the electronic configurations of atoms and ions of the d-block elements of Period 4 (Sc–Zn) given their atomic number and charge (if any).
- Understand why transition metals show variable oxidation numbers.

WHICH ELEMENTS ARE THE TRANSITION METALS?

WHERE ARE THEY FOUND IN THE PERIODIC TABLE?

You will be familiar with the Periodic Table and the numbers of the different groups. Group 1 is on the far left. On the far right are the noble gases, which are in a group numbered 0 (or sometimes 8). When you studied periodicity using the elements in Periods 2 and 3, you ignored the big gap between Groups 2 and 3. However, when you look at Period 4 you can see elements between Groups 2 and 4. Some of these elements are **transition metals**.

You will also be familiar with the idea of classifying elements as belonging to one of the blocks (s, p, d or f) of the Periodic Table. The transition metals are in the d-block of the Periodic Table (**fig A**).
In this topic we will concentrate on the 10 elements in Period 4.

1	2											3	4	5	6	7	0(8)
			H 1														He 2
Li 3	Be 4											B 5	C 6	N 7	O 8	F 9	Ne 10
Na 11	Mg 12											Al 13	Si 14	P 15	S 16	Cl 17	Ar 18
K 19	Ca 20	Sc 21	Ti 22	V 23	Cr 24	Mn 25	Fe 26	Co 27	Ni 28	Cu 29	Zn 30	Ga 31	Ge 32	As 33	Se 34	Br 35	Kr 36
Rb 37	Sr 38	Y 39	Zr 40	Nb 41	Mo 42	Tc 43	Ru 44	Rh 45	Pd 46	Ag 47	Cd 48	In 49	Sn 50	Sb 51	Te 52	I 53	Xe 54
Cs 55	Ba 56	Lanthanide series	Hf 72	Ta 73	W 74	Re 75	Os 76	Ir 77	Pt 78	Au 79	Hg 80	Tl 81	Pb 82	Bi 83	Po 84	At 85	Rn 86
Fr 87	Ra 88	Actinide series															

the d-block elements

▲ **fig A** The position of the transition metals in the periodic table.

WHAT IS A TRANSITION METAL?

It is easier to list the characteristics of transition metals than to explain fully what they are. To describe them as being in the d-block indicates only their location and not their characteristics. Most definitions of a transition metal refer to the electronic configurations of their atoms or ions (see the Subject Vocabulary for a simple definition).

A fuller definition is quoted in the Edexcel specification:

'Transition metals are d-block elements that form one or more stable ions with incompletely filled d-orbitals.'

CHARACTERISTICS OF TRANSITION METALS

Transition metals:

- are hard solids
- have high melting and boiling temperatures
- can act as catalysts
- form coloured ions and compounds
- form ions with different oxidation numbers
- form ions with incompletely filled d-orbitals.

Of the ten d-block elements in Period 4, the first and the last (scandium and zinc) do not have some of these characteristics. So, although it is correct to describe them as d-block elements, they are not considered to be transition metals. In particular, each forms only one ion and their compounds are not coloured.



▲ **fig B** Transition metals include precious metals such as silver and gold, along with familiar metals such as copper, iron and zinc. Each transition metal has more than one oxidation state, and their salts form coloured solutions in water.

ELECTRONIC CONFIGURATIONS

THE ELEMENTS

You learned in **Topic 2 (Book 1: IAS)** how to write the electronic configurations of the elements from hydrogen to krypton, which includes the ten d-block elements in Period 4. For reference, we show these in full in **table A** using the spdf notation, with electrons-in-box diagrams for the electrons in the 4s and 3d orbitals. Copper and chromium do not follow the expected pattern. Where irregularities in the spdf notation exist, they are shown in red. In the electrons-in-box diagrams, paired electrons are shown with a green background and unpaired electrons with a blue background.

It is sometimes acceptable to use an abbreviated shorter form of electronic configuration that represents the complete shells with reference to the previous noble gas. For example, the electronic configuration of titanium is shown in full as $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$, but can be abbreviated as $[Ar]3d^2 4s^2$.

THE IONS

When these elements form ions, they lose one or more electrons. Scandium loses both of its 4s electrons and its only 3d electron, forming the Sc^{3+} ion. Zinc loses both of its 4s electrons and none of its 3d electrons, forming the Zn^{2+} ion. Because these two elements form only one ion each, and these ions have no incompletely filled d-orbitals, they are not classified as transition metals.

LEARNING TIP

Remember that in **Topic 2 (Book 1: IAS)** you learned that the 3d orbital starts to fill before the 4s and that the 4s electrons have higher energy than the electrons in the 3d orbitals. This is why the 4s electrons are lost first when the atoms form ions.

ELEMENT	Z	ELECTRONIC CONFIGURATION	ELECTRONS-IN-BOXES DIAGRAM FOR 3d AND 4s ELECTRONS
scandium	21	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$	↑ ▯ ▯ ▯ ▯ ↑↓
titanium	22	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^2$	↑ ↑ ▯ ▯ ▯ ↑↓
vanadium	23	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$	↑ ↑ ↑ ▯ ▯ ↑↓
chromium	24	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$	↑ ↑ ↑ ↑ ↑ ↑
maganese	25	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$	↑ ↑ ↑ ↑ ↑ ↑↓
iron	26	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7\ 4s^2$	↑↓ ↑ ↑ ↑ ↑ ↑↓
cobalt	27	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$	↑↓ ↑↓ ↑ ↑ ↑ ↑↓
nickel	28	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$	↑↓ ↑↓ ↑↓ ↑ ↑ ↑↓
copper	29	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$	↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑
zinc	30	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2$	↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓

table A Electronic configurations of the elements scandium to zinc.

When the other eight elements form ions, they lose their 4s electrons before their 3d electrons. Each element can lose a variable number of electrons, and so forms ions with different oxidation numbers. The higher oxidation numbers are not found in simple ions. For example, manganese forms an ion with oxidation number +2 (the Mn^{2+} ion), but the manganese ion with oxidation number +7 is MnO_4^-, not Mn^{7+}. Transition metal ions involving higher oxidation numbers usually contain an electronegative element, often oxygen. **Table B** shows the most common oxidation numbers and examples of compounds formed.

ELEMENT	COMMON OXIDATION NUMBERS							COMPOUNDS		
Ti			+3	+4				Ti_2O_3	$TiCl_4$	
V		+2	+3	+4	+5			VCl_3	V_2O_5	
Cr			+3			+6		$CrCl_3$	K_2CrO_4	
Mn		+2		+4			+7	$MnCl_2$	MnO_2	$KMnO_4$
Fe		+2	+3					$FeCl_2$	Fe_2O_3	
Co		+2	+3					$CoSO_4$	$CoCl_3$	
Ni		+2						$NiSO_4$		
Cu	+1	+2						Cu_2O	$CuSO_4$	

table B The most common oxidation numbers and compounds for Period 4 transition elements.

You can see that the highest common oxidation number increases from Ti (+4) to Mn (+7) as all the 4s and 3d electrons become involved in bonding. From Fe to Cu, the increasing nuclear charge means that the electrons are attracted more strongly and are less likely to be involved in bonding. Therefore, ions with higher oxidation numbers are less common.

CHECKPOINT

1. Write out the full and abbreviated electronic configurations for the element chromium.
2. Describe, with reference to its electrons, how an atom of manganese becomes an ion with an oxidation number of +4.

SUBJECT VOCABULARY

transition metal an element that forms one or more stable ions with incompletely filled d-orbitals

17A 2 LIGANDS AND COMPLEXES

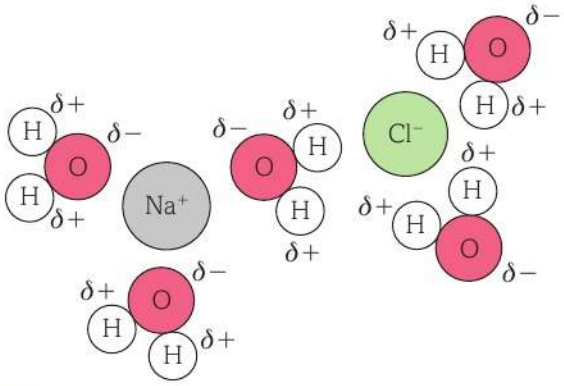
SPECIFICATION REFERENCE
17.4 17.5
17.6 17.10

LEARNING OBJECTIVES

- Know what is meant by the term 'ligand'.
- Understand that dative (coordinate) covalent bonding is involved in the formation of complex ions.
- Know that a complex ion is a central metal ion surrounded by ligands.
- Understand the meaning of the term 'coordination number'.

SYMBOLS AND EQUATIONS WITHOUT LIGANDS

An example of the symbol of a metal ion is Na^+, the sodium ion. You will also have used the symbol $Na^+(aq)$, which shows that the sodium ion is dissolved in water. The process of dissolving solid sodium chloride in water may have been explained in terms of the slightly negative ($\delta-$) oxygen atoms in the water molecules being attracted to the positively charged sodium ions and keeping them in solution (**fig A**). You may have a mental picture of Na^+ being surrounded by an unspecified number of water molecules, with these water molecules leaving and being replaced by others.



▲ **fig A** Positive sodium ions surrounded by $\delta-$ oxygen atoms in water molecules, and negative chloride ions surrounded by $\delta+$ hydrogen atoms in water molecules.

When you write equations for the reactions of transition metal ions you can sometimes use the same method. For example, the iron(II) ion can be shown as Fe^{2+}, or as $Fe^{2+}(aq)$ to show that it is dissolved in water.

The equation for a displacement reaction involving a transition metal ion is:

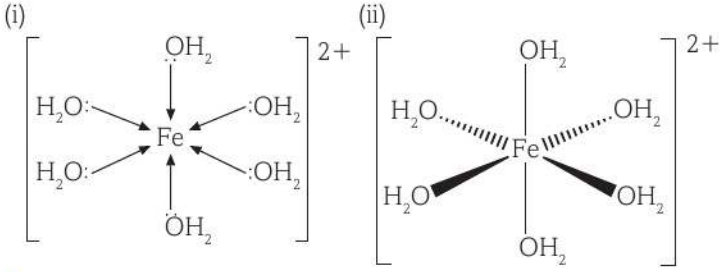
$$Mg(s) + Fe^{2+}(aq) \rightarrow Mg^{2+}(aq) + Fe(s)$$

This equation clearly shows that the reaction is an example of redox, as magnesium transfers electrons to iron(II) ions.

However, it is often better to consider the reactions of transition metal ions in a different way in order to better understand the reactions. This method will be described next.

SYMBOLS AND EQUATIONS WITH LIGANDS

The ions of non-transition metals tend to have larger radii than those of transition metals in the same period of the Periodic Table. For example, the ionic radius of K^+ is 0.133 nm, and that of Fe^{2+} is 0.076 nm. The relatively small size of transition metal ions enables them to attract electron-rich species more strongly, including the water molecules present in aqueous solutions. These water molecules are attracted to the transition metal ions so strongly that they form a specific number of bonds, usually six, giving a structure that can be represented in diagrams such as these:



▲ **fig B** Here are two ways to represent a complex ion: (i) shows one of the lone pairs of electrons on the oxygen atoms and the dative covalent bonds with arrows; (ii) shows the 3D arrangement of bonds and water molecules around the metal ion.

In **fig B** (i):

- the bonds are shown with arrowheads, indicating that they are **dative (coordinate) bonds** – one of the lone pairs of electrons on each oxygen atom is used to form the bond
- the whole structure is shown inside square brackets, and the original charge of the Fe^{2+} ion is shown outside the bracket
- the water molecules are arranged in a regular pattern around the Fe – this arrangement can be explained in terms of the electron pair repulsion theory.

In **fig B** (ii):

- solid wedges represent bonds coming out of the plane of the paper
- striped wedges represent bonds going behind the plane of the paper.

EXAM HINT

Practise drawing these complexes as in **fig B**. Make sure you show which part of the ligand forms the dative bond with the central metal ion. For example, you can represent water as $:OH_2$, to show that the oxygen part of the water molecule is responsible for the bond.

Although it is often useful to think about diagrams like this, when writing equations it is more usual to abbreviate them. In an equation, the example above would be shown as $[Fe(H_2O)_6]^{2+}$. Notice that the square brackets and the position of the charge are the same in this abbreviated form.

These water molecules, and other electron-rich species that can form dative bonds in the same way, are called **ligands**. The complete formulae are called **complexes**. As most complexes have charges, they are often called **complex ions**, but where there is no overall charge, it is better to use the term 'complex'. The total number of dative bonds around the metal ion is called the **coordination number**.

EXAMPLES OF LIGANDS

Many species can act as ligands. However, those that we will meet most often are shown in **table A**, which also shows how the name of the ligand changes when used in the name of a complex.

LIGAND	FORMULA	CHARGE	NAME IN COMPLEX
water	H_2O	0	aqua
hydroxide	^-OH	−1	hydroxo
ammonia	NH_3	0	ammine
chloride	Cl^-	−1	chloro

table A

Note these points about the names used in complexes:

- ligands with a negative charge end in -o
- ammine should not be confused with the organic term amine.

NAMING COMPLEXES

Table B shows some examples to illustrate the naming system used.

COMPLEX	NAME	NOTES
$[Fe(H_2O)_6]^{2+}$	hexaaquairon(II)	In order, there is: • the number of ligands • the name of the ligand • the name of the metal ion • the oxidation number of the metal ion.
$[FeCl_4]^-$	tetrachloroferrate(III)	The overall charge is 1− because the ion is formed from one Fe^{3+} ion and four Cl^- ions. '-ate' is added to the metal name to show that it is now part of a negatively charged complex ion. As it is negatively charged, a Latin name for the metal is used (in Latin, iron is ferrum).
$[Cu(NH_3)_4(H_2O)_2]^{2+}$	tetraamminediaquacopper(II)	With two different ligands, they appear in alphabetical order (so ammine before aqua), ignoring the tetra- and di- parts of the name.

table B

EXAM HINT

In an exam question you may be asked to **name** complex ions.

LEARNING TIP

You can work out the charge on the metal ion in a complex by considering the overall charge on the complex and knowing the charges on the ligands. For example, $[FeCl_4]^{2-}$ has an overall charge of 2−, and there are four chloride ions each with a charge of 1−, so the charge on the metal ion must be 2+.

CHECKPOINT

SKILLS PROBLEM-SOLVING

1. Explain why the methane molecule cannot act as a ligand.
2. An equation for a reaction involving a transition metal complex is:

$$[Cu(H_2O)_6]^{2+} + 2OH^- \rightarrow [Cu(H_2O)_4(OH)_2] + 2H_2O$$

Explain why the first product is a complex but not a complex ion.

SUBJECT VOCABULARY

dative (coordinate) bond a covalent bond formed between the central metal atom or ion and a ligand, in which both of the bonding electrons are supplied by the ligand
ligand a species that uses a lone pair of electrons to form a dative bond with a metal ion
complex a species containing a metal ion joined to ligands
complex ion a complex with an overall positive or negative charge
coordination number the number of dative (coordinate) bonds in the complex

17A 3 THE ORIGIN OF COLOUR IN COMPLEXES

SPECIFICATION REFERENCE
17.7 17.8 17.9

LEARNING OBJECTIVES

- Know that aqueous solutions of transition metal ions are usually coloured.
- Understand that the colour of aqueous ions, and other complex ions, is a consequence of the splitting of the energy levels of the d-orbitals by ligands.
- Understand why there is a lack of colour in some aqueous ions and other complex ions.

A COMPLEX EXPLANATION!

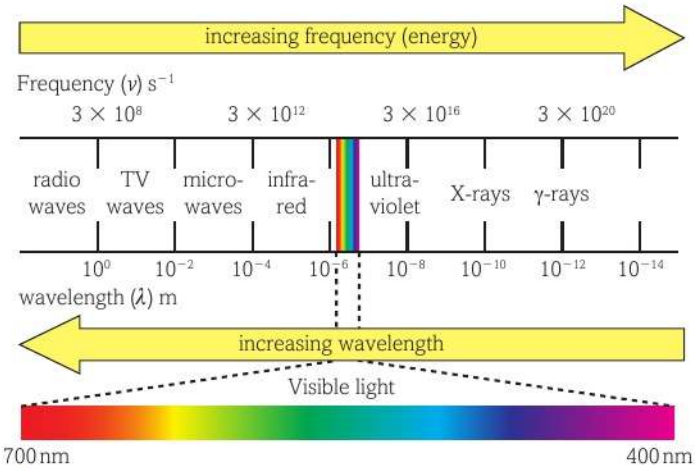
Colour is very important in our world – think of traffic lights, paint charts, clothing and the colours of flowers. There are scientific explanations for colour in plants, paints and dyes. However, these explanations are often specific to the type of chemical substance being considered.

As far as transition metal complexes are concerned, a full explanation includes three main concepts:

- the electromagnetic spectrum and the colour wheel
- the connection between colour, energy and wavelength
- the electronic configurations of transition metal ions.

THE ELECTROMAGNETIC SPECTRUM

We first need to find out about the **electromagnetic spectrum** (**fig A**). You are probably familiar with the idea that the visible part of the electromagnetic spectrum (sometimes described as 'white light') is made up of a mixture of colours (sometimes described as the colours of the rainbow).



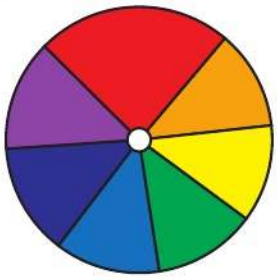
▲ **fig A** The diagram shows how the visible part of the spectrum fits into the complete electromagnetic spectrum.

So, how many colours make up white light? A traditional answer is seven:

red orange yellow green blue indigo violet

However, seven is a randomly chosen number, and this list of colours does not contain some that are now more familiar, such as cyan and magenta.

The real answer is that visible light is made up of a limitless number of colours! If we take red light to have a wavelength of 700 nm and violet light to have a wavelength of 400 nm, then each precise value within this range (such as 651.0 and 438.4) represents a different wavelength and therefore a different colour. There is clearly no limit to the number of values in this range.



▲ **fig B** This is one example of a colour wheel.

THE COLOUR WHEEL

THE COLOURS IN A COLOUR WHEEL

Now consider the colours in the visible spectrum shown as a wheel, with the red and violet ends of the spectrum next to each other. There are many different colour wheels with varying numbers of colours (just try a web search), but **fig B** shows one that contains seven colours corresponding to those listed above. Clockwise from the top, they are shown in order from red to violet.

COMPLEMENTARY COLOURS

In colour wheels, **complementary colours** are the colours shown opposite each other. In this one, red is opposite blue and green. When white light is passed through a solution containing a transition metal complex, some wavelengths of light are absorbed by the complex. The light emerging will therefore contain proportionately more of the complementary colour. So, if a complex absorbs red light, the light emerging will look blue or green (**fig C**).



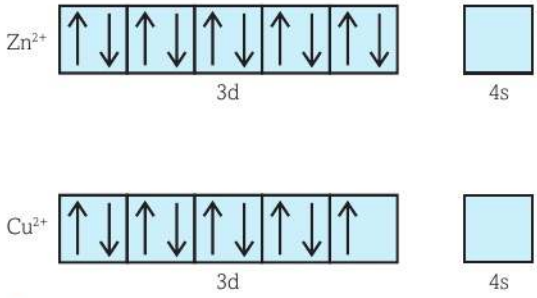
▲ **fig C** A range of transition metal solutions. From left to right: Ti^{2+}, V^{3+}, VO^{2+}, Cr^{3+}, $Cr_2O_7^{2-}$, Mn^{2+}, MnO_4^-, Fe^{3+}, Co^{2+}, Ni^{2+} and Cu^{2+}.

COLOUR DEPENDS ON ELECTRONS IN 3d ENERGY LEVELS

A complete explanation of colour, or absence of colour, in aqueous solutions of metal ions is quite complicated. However, the following explanation should help you to understand the basics. We will use an aqueous solution containing copper(II) ions as an example.

An aqueous solution of zinc sulfate is colourless, but an aqueous solution of copper(II) sulfate is blue. As the sulfate ion is present in both solutions we can assume that any difference in colour is caused by the zinc ions and the copper(II) ions. Zinc and copper are next to each other in the Periodic Table, so why should copper, but not zinc, form coloured ions?

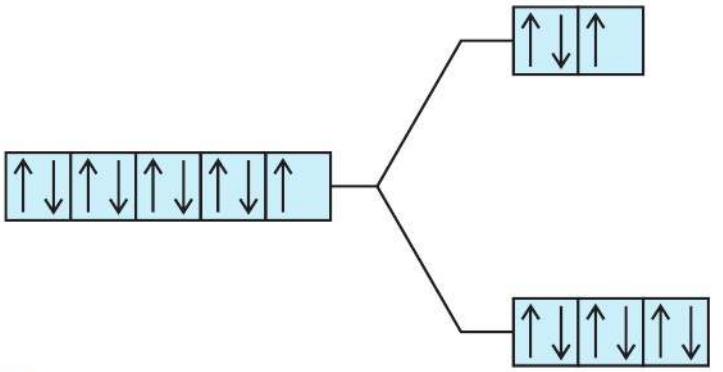
The electronic configurations of the Zn^{2+} and Cu^{2+} ions are shown in **fig D** in the electrons-in-boxes format.



▲ **fig D** The diagram shows the configurations of the 3d and 4s electrons in Zn^{2+} and Cu^{2+}.

Ions that have completely filled 3d energy levels (such as Zn^{2+}) and ions that have no electrons in their 3d energy levels (such as Sc^{3+}) are not coloured. However, the copper(II), or Cu^{2+}, ion has only nine electrons in the 3d energy level, so it is not completely filled. When water ligands are attached

to the copper(II) ion, the energy level splits into two levels with slightly different energies (**fig E**). The lower energy level contains six electrons and the higher energy level contains three electrons.

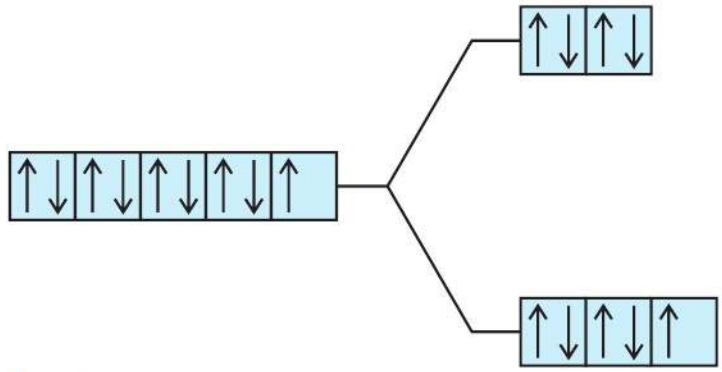


▲ **fig E** In the Cu^{2+} ion, the water ligands split the 3d energy level into lower and higher energy levels.

DID YOU KNOW?

As with a number of theories suggested at advanced levels of the study of chemistry, the concept that colour in transition metal complexes is the result solely of d–d transitions is too simple. The colours of a number of transition metal ions, such as the purple colour of MnO_4^-, are a result of charge-transfer transitions. These transitions occur when an electron moves from a ligand-based orbital to a metal-based orbital, or vice versa. If you are interested, you might like to do some research into this theory.

If one of the electrons in the lower energy level absorbs energy from the visible spectrum, it can move to the higher energy level (**fig F**). Movement from a lower energy level to a higher energy level is called **promotion** or **excitation**. When an electron moves to a higher energy level, the amount of energy it absorbs depends on the difference in energy between the two levels – the bigger the energy difference, the more energy the electron absorbs. It is important to know that the amount of energy gained by the electron is directly proportional to the frequency of the absorbed light (so the energy gained increases as the frequency increases), and inversely proportional to the wavelength of the light (so the energy gained increases as the wavelength decreases). In the case of the Cu^{2+} ion, the small difference in energy levels means that low frequency (or high wavelength) radiation is absorbed from the red end of the spectrum. Therefore, blue light is transmitted.



▲ **fig F** Note that one electron has moved from the lower energy level to the higher energy level.

LEARNING TIP

Try to find a way to remember that red light has a longer wavelength than violet light. The Red Sea (between Africa and Asia) is a long (and narrow) sea. Perhaps you can find a better way!

EXAM HINT

Changing the type of ligand in a complex (for example, from water to ammonia) can change the gap between the split d-orbitals. This results in a change in the colour of the complex.

As an example you could think about the difference in colour between $[Cu(H_2O)_6]^{2+}$ and $[Cu(NH_3)_4(H_2O)_2]^{2+}$ in **Section 17B.1**. Use the colour wheel to decide which colours are being absorbed by each complex.

CHECKPOINT

1. Solutions containing $Fe^{3+}(aq)$ ions are yellow. Explain which part of the electromagnetic spectrum is absorbed by these ions.
2. State why solutions containing $Al^{3+}(aq)$ ions are colourless.

SUBJECT VOCABULARY

electromagnetic spectrum the range of all wavelengths and frequencies of all the types of radiation
complementary colours colours opposite each other on a colour wheel
promotion (or **excitation**) when an electron moves from a lower energy level to a higher energy level

17A 4 COMMON SHAPES OF COMPLEXES

SPECIFICATION REFERENCE 17.13 17.14

LEARNING OBJECTIVES

- Understand why complexes with six-fold coordination have an octahedral shape, such as those formed by metal ions with H_2O, OH^- and NH_3 as ligands.
- Know that transition metal ions may form tetrahedral complexes with relatively large ions such as Cl^-.

PREDICTING THE SHAPES OF COMPLEXES

USING ELECTRON PAIR REPULSION THEORY

You already know from **Topic 3 (Book 1: IAS)** about the use of the electron pair repulsion theory to predict and explain the shapes of simple molecules and ions. It is quite easy to extend this idea to predicting and explaining the shapes of complexes. A ligand bonds to the central metal ion by donating a pair of electrons to form a dative bond. The only differences are that you should ignore the 3d electrons in the transition metal ion and the overall charge on the complex – just count the number of electron pairs donated by the ligands. **Table A** shows how this theory can be applied in exactly the same way to predict the shape of a transition metal complex and the shape of a simple molecule.

NUMBER OF LIGANDS	ELECTRON PAIRS DONATED	SHAPE	BOND ANGLE	EXAMPLE OF A COMPLEX ION	EXAMPLE OF A SIMPLE MOLECULE
6	6	octahedral	90°	$[Co(NH_3)_6]^{2+}$	SF_6
4	4	tetrahedral	109.5°	$[CuCl_4]^{2-}$	CH_4
2	2	linear	180°	$[Ag(NH_3)_2]^+$	$BeCl_2$

table A The electron pair repulsion theory can predict the shapes of simple molecules and complex ions.

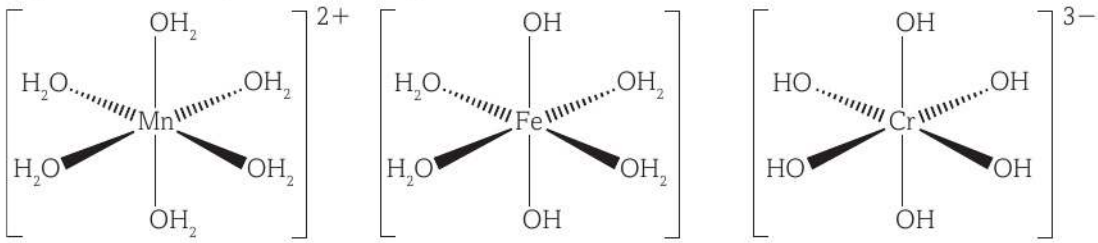
OCTAHEDRAL COMPLEXES

The most common ligands in most octahedral complexes are water, ammonia and the hydroxide ion. These ligands have different numbers of lone pairs of electrons (ammonia has one, water has two and the hydroxide ion has three). However each ligand uses only one lone pair to form a coordinate bond with the transition metal ion. As they contain six ligands, the complexes are sometimes described as having **six-fold coordination**. Note that the electron pair donor in each of these ligands is an element in Period 2 of the Periodic Table, so all three ligands are of approximately equal size. **Table B** shows some common examples.

ABBREVIATED FORMULA	NAME	COLOUR
$[Mn(H_2O)_6]^{2+}$	hexaaquamanganese(II)	very pale pink (usually described as colourless)
$[Fe(H_2O)_4(OH)_2]$	tetraaquadihydroxoiron(II)	pale green
$[Cr(OH)_6]^{3-}$	hexahydroxochromate(III)	green

table B Examples of octahedral complexes.

Fig A shows the shapes of these complexes.



▲ **fig A** The structures of some octahedral complexes.

Note that:

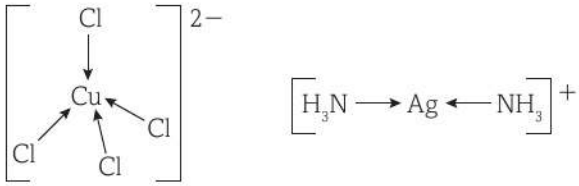
- Solid and striped wedge bonds are used to indicate the shapes, although the 3D shape can be shown in other ways.
- The overall charges are all different – this depends on the charge on the original transition metal ion and on how many negatively charged ligands there are.
- All three structures use square brackets, although these are sometimes left out when the complex is neutral.
- The water ligands are shown as both H_2O and OH_2 – the important point is that the bond must be shown to come from oxygen because it supplies the lone pair of electrons for the dative bond.

TETRAHEDRAL AND LINEAR COMPLEXES

These are much less common than octahedral complexes. The only tetrahedral complexes that you are likely to see in your course are those in which chloride ions act as ligands, such as in the $[CuCl_4]^{2-}$ ion. Note that because chlorine is a Period 3 element, its ions are much bigger than water molecules, ammonia molecules and hydroxide ions. So, there is usually not enough room around the central metal ion for six chloride ions to act as ligands.

The only linear complex you are likely to see in your course is the reactive ion present in Tollens' reagent (sometimes called ammoniacal silver nitrate). An explanation of why the Ag^+ ion has only two ligands, and not six, is beyond the aims of this book, but note that silver is a transition metal in Period 5 (not Period 4) of the Periodic Table. It therefore behaves differently from the transition metals in Period 4.

Fig B shows the shapes of these two complexes.



▲ **fig B** The structures of a tetrahedral complex and a linear complex.

LEARNING TIP

You can usually assume that a complex with six ligands is octahedral, one with four ligands is tetrahedral and one with two ligands is linear.

CHECKPOINT

1. A complex that forms between cobalt ions and nitrite ions (NO_2^-) has the abbreviated formula $[Co(NO_2)_6]^{3-}$. Predict its shape and the oxidation number of the transition metal ion.

2. A complex forms between nickel(II) ions and chloride ions. Predict its shape, name and formula.

SKILLS REASONING

SUBJECT VOCABULARY

six-fold coordination complexes in which there are six ligands forming coordinate bonds with the transition metal ion

17A 5 SQUARE PLANAR COMPLEXES

SPECIFICATION REFERENCE 17.15

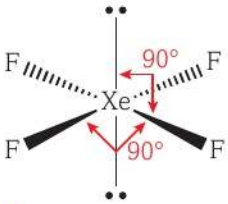
LEARNING OBJECTIVES

- Know that square planar complexes are also formed by transition metal ions and that *cis*-platin is an example of such a complex, which is used in cancer treatment where it is supplied as a single isomer and not in a mixture with the *trans* form.

SQUARE PLANAR MOLECULES

We know that the electron pair repulsion theory can be applied to predict the shapes of simple molecules and ions. However, you may not have seen an example of a **square planar** shape. One example is xenon tetrafluoride, XeF_4. Although xenon is a noble gas, it does form some stable compounds.

An atom of xenon has eight electrons in its outermost energy level, and each fluorine atom uses one of its electrons to form a covalent bond. The outer energy shell now contains 12 electrons. These are arranged in six pairs, forming an octahedral arrangement. Two of these pairs are lone pairs, which repel each other and are therefore located opposite each other. So, the four bonding pairs are in a plane, with the four fluorine atoms at the corners of a square and an F–Xe–F bond angle of 90°, as shown in **fig A**.



▲ **fig A** The square planar shape of the xenon tetrafluoride molecule.

cis-PLATIN

Square planar complexes are much less common than octahedral and tetrahedral complexes. One particular complex of this type, *cis*-platin, has become well known in recent years because of its use as an effective treatment for some types of cancer, especially testicular cancer.

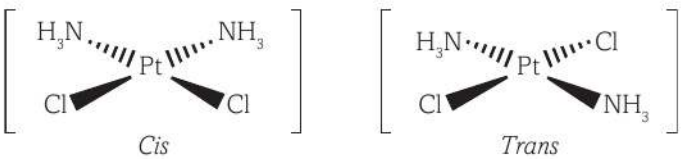
An explanation of why the four ligands in this complex form a square planar shape, and not a tetrahedral shape, is beyond the aims of this book. It is not easily explained by the electron pair repulsion theory. Part of the explanation is that platinum is a transition metal in Period 6, and so behaves differently from the transition metals in Period 4.

CIS-TRANS ISOMERS

You have learned about *E–Z* and *cis-trans* isomerism in **Topic 5 (Book 1: IAS)**. Your understanding of this type of isomerism can now be applied to inorganic compounds such as *cis*-platin and its isomer, *trans*-platin. These consist of a platinum(II) ion, two ammonia ligands and two chloride ion ligands.

Fig B shows the structures of these isomers and the relationship between them.

- The *cis*- prefix indicates that identical ligands are next to each other.
- The *trans*- prefix indicates that they are opposite each other.

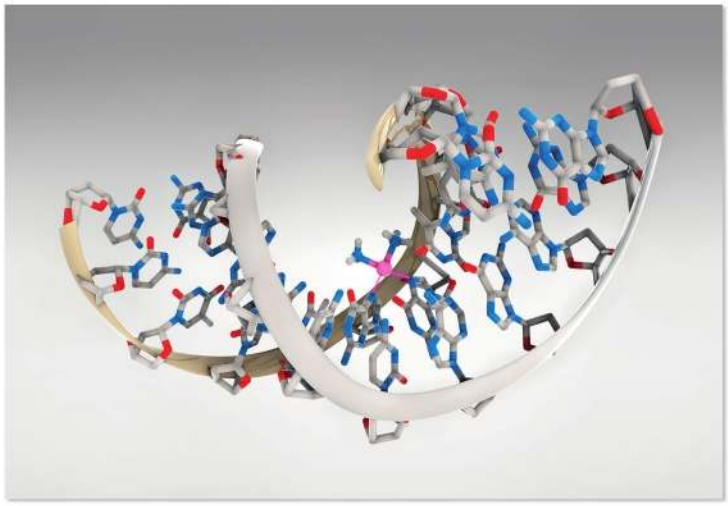


▲ **fig B** Structures of *cis*-platin and *trans*-platin.

ANTI-CANCER ACTION

A full understanding of how *cis*-platin kills cancer cells is beyond the aims of this book. Put simply:

- All cells, including cancer cells, contain deoxyribonucleic acid (DNA).
- During cell division, the two strands of DNA must separate from each other to form more DNA.
- The structure of *cis*-platin enables it to form a bond between the two strands of DNA (**fig C**), which prevents them from separating and so prevents the cancer cells from dividing.



▲ **fig C** DNA strands are shown in grey and *cis*-platin in pink.

trans-PLATIN

The isomers *trans*-platin and *cis*-platin have different structures. The difference in structure means that *trans*-platin is a much less effective cancer treatment than *cis*-platin. It is also more toxic. So, it is important to use only the *cis*- isomer in the cancer treatment.

LEARNING TIP

You should know the structures of *cis*-platin and *trans*-platin, but you do not need to know exactly why *cis*-platin is effective at treating cancer but *trans*-platin is not.

CHECKPOINT

1. Explain why you would expect the ICl_4^- ion to have a square planar shape.
2. (a) Draw the structures of the two *cis-trans* isomers of the $IF_2Cl_2^-$ ion.
   (b) Why are there no *cis-trans* isomers of the $IFCl_3^-$ ion?

SUBJECT VOCABULARY

square planar this shape contains a central atom or ion surrounded by four atoms or ligands in the same plane and with bond angles of 90°

17A 6 MULTIDENTATE LIGANDS

SPECIFICATION REFERENCE
17.12 17.16
17.17 17.25

LEARNING OBJECTIVES

- Understand that H_2O, Cl^- and NH_3 act as monodentate ligands
- Understand the terms 'bidentate' and 'hexadentate' in relation to ligands, and be able to identify examples such as $NH_2CH_2CH_2NH_2$ and $EDTA^{4-}$
- Know that haemoglobin is an iron(II) complex containing a polydentate ligand and that ligand exchange occurs when an oxygen molecule bound to haemoglobin is replaced by a carbon monoxide molecule.
- Understand, in terms of the positive increase in ΔS_{system}, that the substitution of a monodentate ligand by a bidentate or hexadentate ligand leads to a more stable complex ion.

DENTICITY

Denticity is a rarely used English word, but it is a property of ligands that you should be aware of. It comes from the Latin word *dentis* (meaning 'tooth'), from which we get the familiar word 'dentist'. There is a connection!

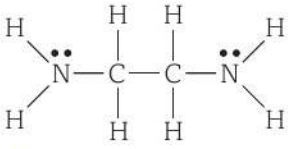
When ligands were introduced earlier in this book we could have described them as **monodentate ligands**. You can think of monodentate as meaning 'one tooth' or 'one bite', which means that the ligand uses one lone pair of electrons on one atom to form the dative bond with the metal ion. H_2O, OH^- and NH_3 all act as monodentate ligands because they use only one pair of electrons to form a dative (coordinate) bond in complexes.

Now for **bidentate ligands.** *Bi* means two, so a bidentate ligand has two atoms, each of which can use a lone pair of electrons to form a dative bond with the metal ion. You can imagine that a bidentate ligand has two atoms that can 'bite' onto the metal ion.

You can work out for yourself what a **multidentate ligand** is – a ligand with several atoms, each of which uses a lone pair of electrons to form a dative bond with the metal ion. The most common example you are likely to meet has six such atoms, so it could be described as a **hexadentate** ligand.

BIDENTATE LIGANDS

The most likely bidentate ligand you will meet in your chemistry course is the organic compound with the structural formula $NH_2CH_2CH_2NH_2$. Its correct name is 1,2-diaminoethane, although for a reason we will soon see, it is sometimes called ethylenediamine – the ethylene comes from the CH_2CH_2 part, and it is correctly described as a diamine because there are two amino groups. Its structure is:



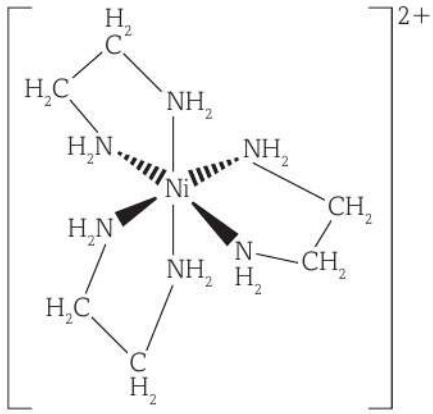
▲ **fig A** The structure of 1,2-diaminoethane.

EXAM HINT

Having two lone pairs is not enough for a molecule to act as a bidentate ligand. N_2H_4 (hydrazine) has two lone pairs. However, it cannot bend to the right shape for both lone pairs to form dative bonds. So N_2H_4 cannot act as a bidentate ligand.

Occasionally it is abbreviated to 'en', especially when used in equations representing its reactions as a ligand.

When it acts as a bidentate ligand, it uses the lone pair of electrons on each nitrogen atom to attach to the metal ion. **Fig B** shows an Ni^{2+} ion joined to three molecules of 1,2-diaminoethane.



▲ **fig B** The structure of the complex formed when Ni^{2+} reacts with three molecules of 1,2-diaminoethane.

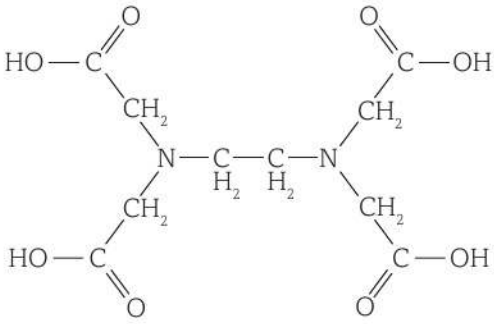
If this complex were formed by the reaction between a hexaaqua metal ion and 1,2-diaminoethane, this abbreviated equation could be written:

$$[Ni(H_2O)_6]^{2+} + 3en \rightarrow [Ni(en)_3]^{2+} + 6H_2O$$

MULTIDENTATE LIGANDS

The most likely multidentate ligand you will see in your chemistry course is an organic ion with a rather complicated structure.

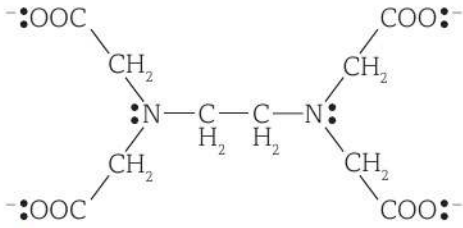
First, consider the structure of 1,2-diaminoethane. Next, imagine that each of the four hydrogen atoms on the two nitrogens are replaced by $-CH_2COOH$ (this is ethanoic acid bonded to the nitrogens via the CH_3 group), as shown in **fig C**.



▲ **fig C** The structure of the EDTA molecule.

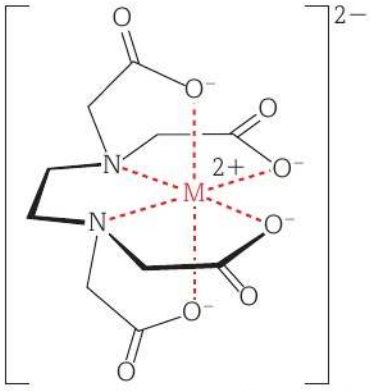
You may know that the old name for ethanoic acid is acetic acid. Four of these molecules have been used to form this structure. Now, remembering the alternative name for 1,2-diaminoethane, you should be able to see that one name for this structure could be ethylenediaminetetraacetic acid. This is difficult to say, so pick out four key letters from the name to get the abbreviation that is normally used for the compound. EthyleneDiamineTetraAcetic acid now becomes 'EDTA'. We will not attempt the IUPAC name!

The final step is to consider the ion formed when each of the ethanoic acid groups loses its H^+ ion – this gives an ion with four negative charges. It is a hexadentate ligand referred to as $EDTA^{4-}$ and has this structure:



▲ **fig D** The structure of the $EDTA^{4-}$ ion.

The advantage of displaying the $EDTA^{4-}$ ion in this way is that it clearly shows the six lone pairs of electrons that are used to form the dative bonds when the ion acts as a ligand (two pairs on the nitrogen atoms and four pairs on the oxygen atoms). **Fig E** uses a skeletal formula to show the complex formed between one M^{2+} ion and one $EDTA^{4-}$ ion.



▲ **fig E** The structure of the complex formed between M^{2+} and $EDTA^{4-}$.

The six dative bonds formed by $EDTA^{4-}$ are shown as dashed lines.

THE STABILITY OF COMPLEXES

Most complexes are stable, in the sense that they do not decompose easily. In this heading, 'stability' does not refer to complexes containing transition metal ions with unstable oxidation numbers, such as +2 in scandium (Sc^{2+}). Instead, it refers to a comparison of the stabilities of two complexes in which the number of ligands has changed. For a discussion of 'stability' see **Topic 12**. Consider this ligand exchange reaction in which a monodentate ligand is replaced by a bidentate ligand:

$$[Cu(H_2O)_6]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 6H_2O$$

Six water ligands are replaced by three 1,2-diaminoethane ligands, so the total number of species has increased from four to seven. This means that the system is more disordered, and so there is an increase in ΔS_{system}. Ligand exchange reactions of this sort lead to an increase in stability of the products compared to the reactants, and so formation of the products is favoured.

When a monodentate ligand is replaced by a bidentate or hexadentate ligand, the increase in stability is even greater, as in this example with $EDTA^{4-}$:

$$[Cu(H_2O)_6]^{2+} + EDTA^{4-} \rightarrow [Cu(EDTA)]^{2-} + 6H_2O$$

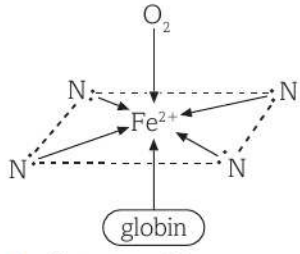
Here, six water ligands are replaced by one $EDTA^{4-}$ ligand, so the total number of species has increased from two to seven.

HAEMOGLOBIN AND OXYGEN TRANSPORT

You may know that haemoglobin is a protein in red blood cells that plays a vital role in transporting oxygen through the bloodstream in humans and other animals. Here is a simplified explanation of the role of haemoglobin in oxygen transport, without including details of its structure and of the haem group that it contains:

- Haemoglobin consists of three main parts, the largest of which is protein (the 'globin' part).
- Within the protein, there are four haem groups that are made up mostly of carbon and hydrogen atoms.
- Inside each haem group, there are four nitrogen atoms that hold an Fe^{2+} ion by forming dative bonds with it in a square planar structure.
- There is a fifth dative bond from the protein to the Fe^{2+} ion.
- When blood passes through the lungs, haemoglobin collects oxygen molecules and transports them to cells, where it is released.
- When haemoglobin collects oxygen, the oxygen molecule acts as a ligand by using one of its lone pairs of electrons to form a dative bond with one of the Fe^{2+} ions.

Fig F is a very simplified diagram showing the six dative bonds in oxyhaemoglobin.



▲ **fig F** Part of the structure of oxyhaemoglobin, showing the dative covalent bonds to the central metal ion.

HAEMOGLOBIN AND CARBON MONOXIDE

A carbon monoxide molecule has a lone pair of electrons on its carbon atom that enables it to act as a ligand. The strength of the dative bond between oxygen and haemoglobin is not particularly strong. However, the advantage of this is that the oxygen molecule is relatively easily released when it is needed.

Unfortunately, a much stronger dative bond forms between carbon monoxide and haemoglobin. This means that any carbon monoxide breathed in is very likely to replace the oxygen already bound to haemoglobin – this is a ligand substitution reaction. Once the carbon monoxide has formed carboxyhaemoglobin, the dative bond is so strong that it does not break easily. This means that if enough haemoglobin molecules are converted to carboxyhaemoglobin, there may be too little oxygen transported to support life.

Put very simply, this reaction is reversible:

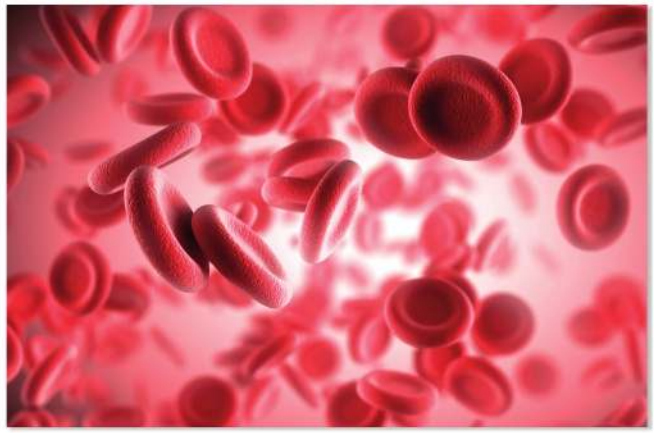
haemoglobin + oxygen $\rightleftharpoons$ oxyhaemoglobin

but this reaction is not easily reversible

haemoglobin + carbon monoxide $\rightarrow$ carboxyhaemoglobin

DID YOU KNOW?

Carbon monoxide poisoning can be treated by breathing in pure oxygen to reverse the process of absorption of carbon monoxide. In severe cases of carbon monoxide poisoning, a complete blood transfusion is required.



▲ **fig G** These red blood cells contain haemoglobin that transports oxygen from the lungs to the cells in the body.

LEARNING TIP

In 1,2-diaminoethane, there are only two lone pairs of electrons in the molecule and both are used when it acts as a ligand. In $EDTA^{4-}$ the lone pairs of electrons on the O of the C=O are not used. Only those on the N atoms and the negatively charged O atoms are used.

CHECKPOINT

1. The water molecule contains three atoms and has two lone pairs of electrons. Explain why it can only act as a monodentate ligand, and not as a bidentate ligand.
2. Write an equation for the ligand substitution reaction that occurs when four of the ligands in the hexaaquairon(III) ion are replaced by 1,2-diaminoethane (en).

SKILLS INTELLECTUAL INTEREST AND CURIOSITY

SUBJECT VOCABULARY

monodentate ligand a molecule or ion that forms one dative bond with a metal ion
bidentate ligand a molecule or ion that forms two dative bonds with a metal ion
multidentate ligand a molecule or ion that forms several dative bonds with a metal ion
hexadentate ligand a molecule or ion that forms six dative bonds with a metal ion